Since diagonal elements of $I–B$ are positive and $|I-B|$ is positive,

Hawkins – Simon conditions are satisfied

Hence this system has a solution.

Example 1.25

In an economy there are two industries P_1 and P_2 and the following table gives the supply and the demand position in crores of rupees.

Production sector	Consumption sector P_1	Consumption sector P_2	Final demand	Gross output
P_1	10	25	15	50
P_2	20	30	10	60

Determine the outputs when the final demand changes to 35 for P_1 and 42 for P_2.

Solution

$a_{11} = 10 \quad a_{12} = 25 \quad x_1 = 50$

$a_{21} = 20 \quad a_{22} = 30 \quad x_2 = 60$

$b_{11} = \frac{a_{11}}{x_1} = \frac{10}{50} = \frac{1}{5};$

$b_{12} = \frac{a_{12}}{x_2} = \frac{25}{60} = \frac{5}{12}$

$b_{21} = \frac{a_{21}}{x_1} = \frac{20}{50} = \frac{2}{5};$

$b_{22} = \frac{a_{22}}{x_2} = \frac{30}{60} = \frac{1}{2}$

The technology matrix is $B = \begin{bmatrix} \frac{1}{5} & \frac{5}{12} \\ \frac{2}{5} & \frac{1}{2} \end{bmatrix}$

$$I - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{5} & \frac{5}{12} \\ \frac{2}{5} & \frac{1}{2} \end{bmatrix}$$

$= \begin{bmatrix} \frac{4}{5} & -\frac{5}{12} \\ -\frac{2}{5} & \frac{1}{2} \end{bmatrix}$, elements of main diagonal are positive

$$\text{Now, } |I-B| = \begin{vmatrix} \frac{4}{5} & -\frac{5}{12} \\ -\frac{2}{5} & \frac{1}{2} \end{vmatrix}$$

$$= \frac{7}{30} > 0$$

Main diagonal elements of $|I-B|$ are positive and $|I-B|$ is positive

$\therefore$ The problem has a solution

$$adj\,(I–B) = \begin{bmatrix} \frac{1}{2} & \frac{5}{12} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix}$$

$$(I–B)^{-1} = \frac{1}{|I-B|} adj(I-B)$$

(Since $|I-B| \neq 0, (I-B)^{-1}$ exist)

$$= \frac{1}{\frac{7}{30}} adj(I-B)$$

$$= \frac{30}{7}\begin{bmatrix} \frac{1}{2} & \frac{5}{12} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 15 & \frac{25}{2} \\ 12 & 24 \end{bmatrix}$$

$X = (I–B)^{-1}D$, where $D = \begin{bmatrix} 35 \\ 42 \end{bmatrix}$

$$= \frac{1}{7}\begin{bmatrix} 15 & \frac{25}{2} \\ 12 & 24 \end{bmatrix}\begin{bmatrix} 35 \\ 42 \end{bmatrix} = \begin{bmatrix} 150 \\ 204 \end{bmatrix}$$

The output of industry P_1 should be ₹150 crores and P_2 should be ₹204 crores.

Example 1.26

An economy produces only coal and steel. These two commodities serve as intermediate inputs in each other's production. 0.4 tonne of steel and 0.7 tonne of coal are needed to produce a tonne of steel. Similarly 0.1 tonne of steel and 0.6 tonne of coal are required to produce a tonne of coal. No capital inputs are needed. Do you think that the system is viable? 2 and 5 labour days are required to produce a tonnes of coal and steel

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